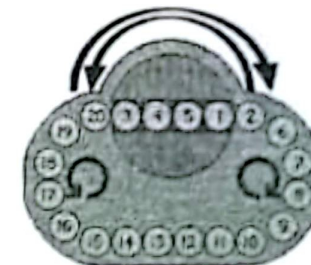
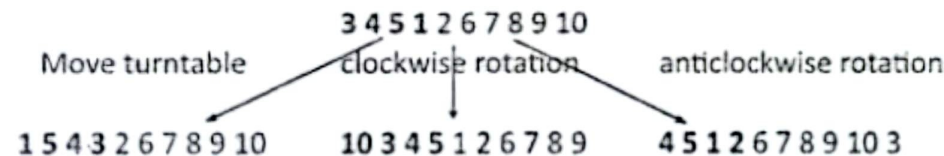




STUDENT ID: _

Course: Artificial Intelligence	Search Problems	Time Allowed: 40min	10
Course Code: AI2002		Date: 02/16/2026	

Q2. The Oval Track puzzle consists of 20 numbered round pieces in one long looped track. Users can slide all the pieces around the loop or use the turntable in the loop which can rotate any four adjacent pieces so that they will be in reverse order. The goal is to arrange the pieces in numerical order from some jumbled-up sequence. Consider the Oval track puzzle with only 10 numbered round pieces. The tree shows the initial state and next actions with their corresponding states (up to level 1). The state is represented by a string of numbers starting with 4 numbers on the turntable. Imagine the last number is circling back to the first number. E.g. in the initial state the number 10 is adjacent to the number 3.



- A. i. Complete the search tree up to level 3.
How many nodes of this search tree are visited if using ii. BFS iii. Depth Limited DFS with a limit of 3
Justify your answer by showing working and giving the order of states visited.
- B. i. Suggest a suitable heuristic for the Oval Track Puzzle and show that your heuristic is admissible
Utilizing your heuristic how many nodes are visited if using: ii. A* Search iii. Greedy first Search
Justify your answer by showing working and giving the order of states visited.

(i)

3 4 5 1 2 6 7 8 9 10

S_0
↓ anti clockwise
 S_1
7 8 9 10 3